CLASSIFICATION

Text Classification

Performance measures: Precision, Recall, and the F measure

Gold standards

- In spam detection (for example) for each item (email document)
  - we therefore need to know whether our system called it spam or not
  - We also need to know whether the email is actually spam or not, i.e. the human-defined labels for each document
  - We will refer to these human labels as the **gold labels**.

Gold Labels, Annotators and Agreement

- Multiple annotators

Utterances	Ann1	Ann2	Raw Agreement	Ann Rand	Agreement (A1, Rand)	Agreement (A2, Rand)
S1	+	+	1	-	0	0
S2	-	-	1	+	0	0
S3	+	-	0	+	1	0
S4	-	+	0	-	1	0
…	…	…	…	…	…	…
			Sum of Agreements		Sum of chance agreements	Sum of chance agreements

- Raw agreement: total number of agreed-upon annotations divided by the total number of possible annotations
- Chance agreement:
- Cohen's Kappa and Krippendorff's alpha
- Agreement over a random subset with an expert annotator

Gold Labels, Annotators and Agreement

- Chance Agreement

Example	Annotator 1	Annotator 2
1	Joy	Joy
2	Sadness	Joy
3	Anger	Anger
4	Fear	Fear
5	Joy	Sadness

- Let's consider an example of inter-annotator agreement for the task of **identifying emotions in text**. Suppose we have two annotators and five examples of text to annotate.

The frequency of each annotation is:

- Joy: 4, Sadness: 2 Anger: 2, Fear: 2
- Calculate the total number of possible annotations. Since we have two annotators and five examples, the total number of possible annotations is:

Total Number of Possible Annotations = 2 x 5 = 10

Now we can calculate the chance agreement using the formula :

Chance Agreement = Sum of (Frequency of Each Annotation / Total Number of Possible Annotations)^2

(3/10)^2 + (2/10)^2 + (2/10)^2 + (2/10)^2 = ?

Cohen's Kappa

Cohen's Kappa Statistic is used to measure the level of agreement between two raters or judges who each classify items into mutually exclusive categories.

$$k = (p_o - p_e) / (1 - p_e)$$

where:

- p_o: Relative observed agreement among raters
- p_e: Hypothetical probability of chance agreement

https://www.statology.org/cohens-kappa-statistic/

Cohen's Kappa

Cohen's Kappa always ranges between 0 and 1, with 0 indicating no agreement between the two raters and 1 indicating perfect agreement between the two raters.

Suppose two museum curators are asked to rate 70 paintings on whether they're good enough to be hung in a new exhibit.
The following 2×2 table shows the results of the ratings:

		Rater 2	
		Yes	No
Rater 1	Yes	25	10
	No	15	20

Cohen's Kappa

Step 1: Calculate relative agreement (p_o) between raters

the proportion of total ratings that the raters both said "Yes" or both said "No" on. We can calculate this as:

- p_o = (Both said Yes + Both said No) / (Total Ratings)
- p_o = (25 + 20) / (70) = **0.6429**

Step 2: Calculate the hypothetical probability of chance agreement (p_e) between raters

Next, calculate the probability that the raters could have agreed purely by chance.

This is calculated as the total number of times that Rater 1 said "Yes" divided by the total number of responses, multiplied by the total number of times that Rater 2 said "Yes" divided by the total number of responses, added to the total number of times that Rater 1 said "No" multiplied by the total number of times that Rater 2 said "No." For our example, this is calculated as:

- P("Yes") = ((25+10)/70) * ((25+15)/70) = 0.285714
- P("No") = ((15+20)/70) * ((10+20)/70) = 0.214285
- p_e = 0.285714 + 0.214285 = **0.5**

Cohen's Kappa

Step 3: Calculate Cohen's Kappa

Lastly, we'll use p_o and p_e to calculate Cohen's Kappa:

- $k = (p_o - p_e) / (1 - p_e)$
- $k = (0.6429 - 0.5) / (1 - 0.5)$
- $k = 0.2857$

Cohen's Kappa turns out to be **0.2857**.

Cohen's Kappa	Interpretation
0	No agreement
0.10 - 0.20	Slight agreement
0.21 - 0.40	Fair agreement
0.41 - 0.60	Moderate agreement
0.61 - 0.80	Substantial agreement
0.81 - 0.99	Near perfect agreement
1	Perfect agreement

- Imagine you're the CEO of the *Delicious Pie Company* and you need to know what people are saying about your pies on social media
- You build a system that detects tweets concerning Delicious Pie
  - the positive class is tweets about Delicious Pie
  - the negative class is all other tweets.
- Imagine that we looked at a million tweets
  - only 100 of them are discussing their love (or hatred) for our pie
  - the other 999,900 are tweets about something completely unrelated
  - **Imagine a simple classifier that stupidly classified every tweet as "not about pie"**
  - This classifier would have 999,900 true positives and only 100 false negatives for an accuracy of 999,900/1,000,000 or 99.99%!
- Accuracy is not a good metric when the goal is to discover something that is rare, or at least not completely balanced in frequency
- A very common situation in the world.

Accuracy $= \frac{My\ Correct\ Answers}{All\ Questions} = \frac{tp + tn}{tp+tn+fp+fn}$

(What fraction of time am I correct in my classification)

Precision $= \frac{True\ Positives}{My\ Positives} = \frac{tp}{tp+fp}$

(How much should you trust me when I say that something tests positive OR what fraction of my positives are true positives)

Recall = Sensitivity $= \frac{True\ Positives}{Real\ Positives} = \frac{tp}{tp+fn}$

(How much of the reality has been covered by my positive output? OR what fraction of the true positives is captured by my positives)

The recall for the “nothing is pie” classifier is 0

Assume:

- 10 gold and 11 copper coins are hidden in the room
- Your task to find me all gold coins and reject copper ones
- Suppose:
  - You return with 7 coins (these are your positives)
    - 5 of these are gold (these are your true positives, tp)
    - 2 are copper (these are your false positives, fp)
    - You left out 5 gold ones (false negatives, fn)
    - You correctly rejected 9 copper ones (true negatives, tn)
    - Your precision is tp/all your positives = tp/(tp+fp) = 5/7
    - Your recall is tp/all real positives = tp/number of actual gold coins = tp/(tp+fn) = 5/10
    - Your accuracy is correct answers/all attempts = (tp+tn)/(tp+tn+fp+fn) = (5+9)/(5+9+2+5) = 14/21
- Extremes:
  - You come back with only one coin and that is gold.
    - Your precision is very high but recall is very low
  - You return all 21 coins
    - Your recall is very high but precision very low
- We need to look at both precision and recall to find a good balanced classifier score

The 2-by-2 contingency table

	correct	not correct
selected	tp	fp
not selected	fn	tn

Precision and recall

- **Precision**: % of selected items that are correct
  Recall: % of correct items that are selected

		gold standard labels		
		gold positive	gold negative	
system output labels	system positive	**true positive**	**false positive**	**precision** $= \frac{tp}{tp+fp}$
	system negative	**false negative**	**true negative**	
		recall $= \frac{tp}{tp+fn}$		**accuracy** $= \frac{tp+tn}{tp+fp+tn+fn}$

Figure 4.4 Contingency table

A combined measure of Precision and Recall

- It is useful to have a single number to describe performance
- The mean of precision and recall?
- But what kind of a mean should we use?

Means

- **Arithmetic Mean**

$$AM = \frac{a_1+a_2+a_3+\cdots+a_n}{}$$

For 2 values: $$AM = \frac{a_1+a_2}{2}$$

- **Geometric Mean**

$$GM = \sqrt[n]{a_1 + a_2 + a_3 + \cdots + a_n}$$

For 2 values: $$GM = \sqrt[2]{a_1 + a_2}$$

- **Harmonic Mean**

$$HM = \frac{}{\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{a_3}+\cdots+\frac{1}{a_n}}$$

For 2 values: $$HM = \frac{2}{\frac{1}{a_1}+\frac{1}{a_2}} = \frac{2a_1a_2}{a_1+a_2}$$

Properties

Ref: http://economistatlarge.com/finance/applied-finance/differences-arithmetic-geometric-harmonic-means

Arithmetic Mean

- The most common average
- Simplest to compute
- The AM of 2, 3 and 4 is 3
- Suitable when:
  - the data is not skewed (no extreme outliers)
  - the individual data points are not dependent on each other

Harmonic Mean

- The harmonic mean of a set of numbers is the reciprocal of the arithmetic mean of reciprocals
- Best used in situations where extreme outliers exist in the population
  - E.g. data: 35, 48, 35, 40, 50, 35, 35, 40, **150**, 35, 40, 35, 45, 45
  - AM = 47.7, HM = 41

Geometric Average

- Geometric mean should be used whenever the data points are inter-related

A combined measure

- Use harmonic mean instead of arithmetic mean as we are taking the average of ratios
- Harmonic mean *punishes* extreme values more strictly:
  - Consider a *trivial* classifier (always returns class A)
  - A very large number of data elements of class B, and a single element of class A:
    - Precision: 0.0
    - Recall: 1.0
    - Arithmetic mean = 0.5 (50% correct), despite being the *worst* possible outcome!
    - The harmonic mean is nearly 0.
    - To have a high F-score, you need *both* a high precision and high recall

F-MEASURE

- A combined measure that assesses the P/R tradeoff is F measure (weighted harmonic mean):

$$F = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2PR}{P + R}$$

- We can choose to favor precision or recall by using an interpolation weight *α:*

$$F = \frac{1}{\alpha \frac{1}{P} + (1-\alpha) \frac{1}{R}}$$

$$= \frac{(\beta^2+1)PR}{\beta^2 P + R} \qquad \beta = \frac{1-\alpha}{\alpha}$$

- Balanced F1 measure has $\beta = 1$ (that is, $\alpha = ½$) as shown above.

By using the harmonic mean in the denominator of the F-measure formula, the F-measure gives equal weight to precision and recall, even when they are very different.

F-MEASURE

$$F_\beta = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

- The β parameter differentially weighs the importance of recall and precision
  - F2 measure, where recall is weighed more heavily
  - F0.5 measure, precision is weighed more heavily

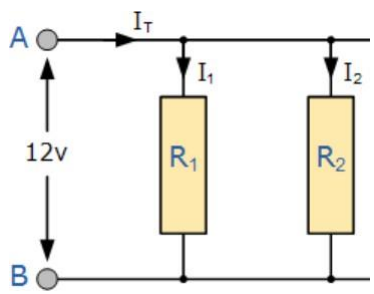
- When $\beta = 1$, precision and recall are equally balanced
- This is the most frequently used metric, and is called F_β=1 or just $F1$:

$$F_1 = \frac{2PR}{P + R}$$

Intuition

Ref: https://stats.stackexchange.com/questions/49226/how-to-interpret-f-measure-values, http://www.electronics-tutorials.ws/resistor/res_4.html

- The formula for F-measure (F1, with β=1) is the same as the formula giving the equivalent resistance composed of two resistances placed in parallel in physics (forgetting about the factor 2).



$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

- This analogy would define F-measure as the equivalent resistance formed by *sensitivity* and *precision* placed in parallel.
- Net resistance gets reduced as soon as any one among the two looses resistance

More than two classes

More than two classes

- Lots of classification tasks in language processing have more than two classes:
  - Sentiment analysis (positive, negative, neutral),
  - Part-of-speech tagging (|POS tags|)
  - Word sense disambiguation (|word senses|)
  - Emotion detection (|emotions|)

More Than Two Classes: Sets of binary classifiers

- Any-of or multi-label classification
  - A document can belong to 0, 1, or more than one classes.
- For each class $c \in C$
  - Build a binary classifier γ_c to distinguish c from all other classes $c' \in C$ (c vs not c)
- Given test doc d,
  - Evaluate it for membership in each class using each γ_c
  - d belongs to any class for which γ_c returns true

More Than Two Classes: Sets of binary classifiers

- One-of or multinomial classification
  - Classes are mutually exclusive: each document in exactly one class
- For each class $c \in C$
  - Build a classifier γ_c to distinguish c from all other classes $c' \in C$.
- Given test doc d,
  - Evaluate it for membership in each class using each γ_c
  - d belongs to the one class with maximum score

Evaluation

- 3-way ***one-of*** email categorization decision (urgent, normal, spam)

		gold labels			
		urgent	normal	spam	
system output	urgent	8	10	1	$\textbf{precision}_u = \frac{8}{8+10+1}$
	normal	5	60	50	$\textbf{precision}_n = \frac{60}{5+60+50}$
	spam	3	30	200	$\textbf{precision}_s = \frac{200}{3+30+200}$
		$\textbf{recall}_u = \frac{8}{8+5+3}$	$\textbf{recall}_n = \frac{60}{10+60+30}$	$\textbf{recall}_s = \frac{200}{1+50+200}$	

Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2

Per class evaluation measures

Recall:

Fraction of docs in class *i* classified correctly:

$$\frac{c_{ii}}{\sum_j c_{ij}}$$

Precision:

Fraction of docs assigned class *i* that are actually about class *i*:

$$\frac{c_{ii}}{\sum_j c_{ji}}$$

Accuracy: (1 - error rate)

Fraction of docs classified correctly:

$$\frac{\sum_i c_{ii}}{\sum_j \sum_i c_{ij}}$$

Micro- vs. Macro-Averaging

- If we have more than one class, how do we combine multiple performance measures into one quantity?
- **Macro-averaging**: Compute performance for each class, then average.
- **Micro-averaging**: Collect decisions for all classes, compute FP and FN for each class.

Evaluation

Macro-average Precision

system output \ gold labels	urgent	normal	spam	
urgent	8	10	1	$\text{precision}_u = \frac{8}{8+10+1}$
normal	5	60	50	$\text{precision}_n = \frac{60}{5+60+50}$
spam	3	30	200	$\text{precision}_s = \frac{200}{3+30+200}$
	$\text{recall}_u = \frac{8}{8+5+3}$	$\text{recall}_n = \frac{60}{10+60+30}$	$\text{recall}_s = \frac{200}{1+50+200}$	

Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2

Class 1: Urgent

	true urgent	true not
system urgent	8	11
system not	8	340

$$\text{precision} = \frac{8}{8+11} = .42$$

Class 2: Normal

	true normal	true not
system normal	60	55
system not	40	212

$$\text{precision} = \frac{60}{60+55} = .52$$

Class 3: Spam

	true spam	true not
system spam	200	33
system not	51	83

$$\text{precision} = \frac{200}{200+33} = .86$$

$$\text{macroaverage precision} = \frac{.42+.52+.86}{3} = \mathbf{.60}$$

The **weighted-averaged** F1 score is calculated by taking the mean of all per-class F1 scores

Evaluation

Micro-average Precision

system output \ gold labels	urgent	normal	spam	
urgent	8	10	1	$\text{precision}_u = \frac{8}{8+10+1}$
normal	5	60	50	$\text{precision}_n = \frac{60}{5+60+50}$
spam	3	30	200	$\text{precision}_s = \frac{200}{3+30+200}$
	$\text{recall}_u = \frac{8}{8+5+3}$	$\text{recall}_n = \frac{60}{10+60+30}$	$\text{recall}_s = \frac{200}{1+50+200}$	

Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes $\langle c_1, c_2 \rangle$, how many documents from c_1 were (in)correctly assigned to c_2

Micro-avg precision= TP1+TP2+TP3/ TP1+TP2+TP3+FP1+FP2+FP3

TP1=8, TP2= 60, TP3=200

FP1= 10+1

FP2= 50+5

FP3= 30 + 3

Micro-avg Precision= 268/268+99=0.73

Evaluation

Micro-average F1 measure

system output \ gold labels	urgent	normal	spam	
urgent	8	10	1	$\text{precision}_u = \frac{8}{8+10+1}$
normal	5	60	50	$\text{precision}_n = \frac{60}{5+60+50}$
spam	3	30	200	$\text{precision}_s = \frac{200}{3+30+200}$
	$\text{recall}_u = \frac{8}{8+5+3}$	$\text{recall}_n = \frac{60}{10+60+30}$	$\text{recall}_s = \frac{200}{1+50+200}$	

Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2

TP= TP1=8 + TP2= 60 + TP3=200

TP=268

FP= FP1= 10+1 + FP2= 50+5 + FP3= 30 + 3

FP= 99

FN=5+3 + 10+30 + 1+50 = 99

$$\frac{TP}{TP+\frac{1}{2}(FP+FN)}$$

Micro-avg Fmeasure = 0.73

Evaluation

- A micro-average is dominated by the more frequent class (in this case spam)
  - as the counts are pooled
- The macro-average better reflects the statistics of the smaller classes,
  - is more appropriate when performance on all the classes is equally important.

Weighted Average that will further weight each class w.r.t to its instances.

Evaluation

- In general, if you are working with an imbalanced dataset where all classes are equally important, using the **macro** average would be a good choice as it treats all classes equally.

- If you have an imbalanced dataset but want to assign greater contribution to classes with more examples in the dataset, then the weighted average is preferred.
  - This is because, in weighted averaging, the contribution of each class to the F1 average is weighted by its size.

Test sets and cross-validation

- We use:
  - the **training set** to train the model,
  - the **development test set** (also called a **devset**) to perhaps tune some parameters and decide what the best model is
  - Run the best model on **<u>unseen</u> test set** to report its performance (precision, recall, F-measure, accuracy, error rate)
- The use of a devset avoids overfitting the to test set
- But having a fixed training set, devset, and test set creates another problem:
  - in order to save lots of data for training, the test set (or devset) might not be large enough to be representative
  - It would be better if we could somehow use **all** our data both for training and test.
- We do this by **cross-validation**:
  - Multiple tain test pairs
  - Randomly choose a training and test set division of our data, train our classifier, and then compute the error rate on the test set
  - Then repeat with a different randomly selected training set and test set.
  - We do this sampling process 10 times and average these 10 runs to get an average error rate.
  - This is called **10-fold cross-validation**
  - Keep one accuracy result, or take mean or std deviation of the k folds
  - Reporting values over multiple samples either through cross-validation preferred for robust evaluation

Cross-validation

- The only problem with cross-validation is:
  - Because all the data is used for testing, we need the whole corpus to be blind i.e. we can't examine any of the data to suggest possible features
  - But looking at the corpus is often important for designing the system
- It is common to create a fixed training set and test set, then do 10-fold cross-validation **inside the training set**, but compute error rate the normal way in the test set
- Comparing two classifiers
  - Cross validate with classifier A and B separately
  - Report their mean and std deviation
  - Resample the data multiple times for each classifier

Cross-validation

- Comparing two trained classifiers
  - When you only have test set
  - See the difference of the two classifiers
  - Acc(test, A) – Acc(test, B)= 0.2
  - A performs better then B
  - Resample the test set multiple times and then report the confidence interval
    - Resampling with replacement
    - If A performs better than B 70 times then report the results

Cross-validation with fixed test data

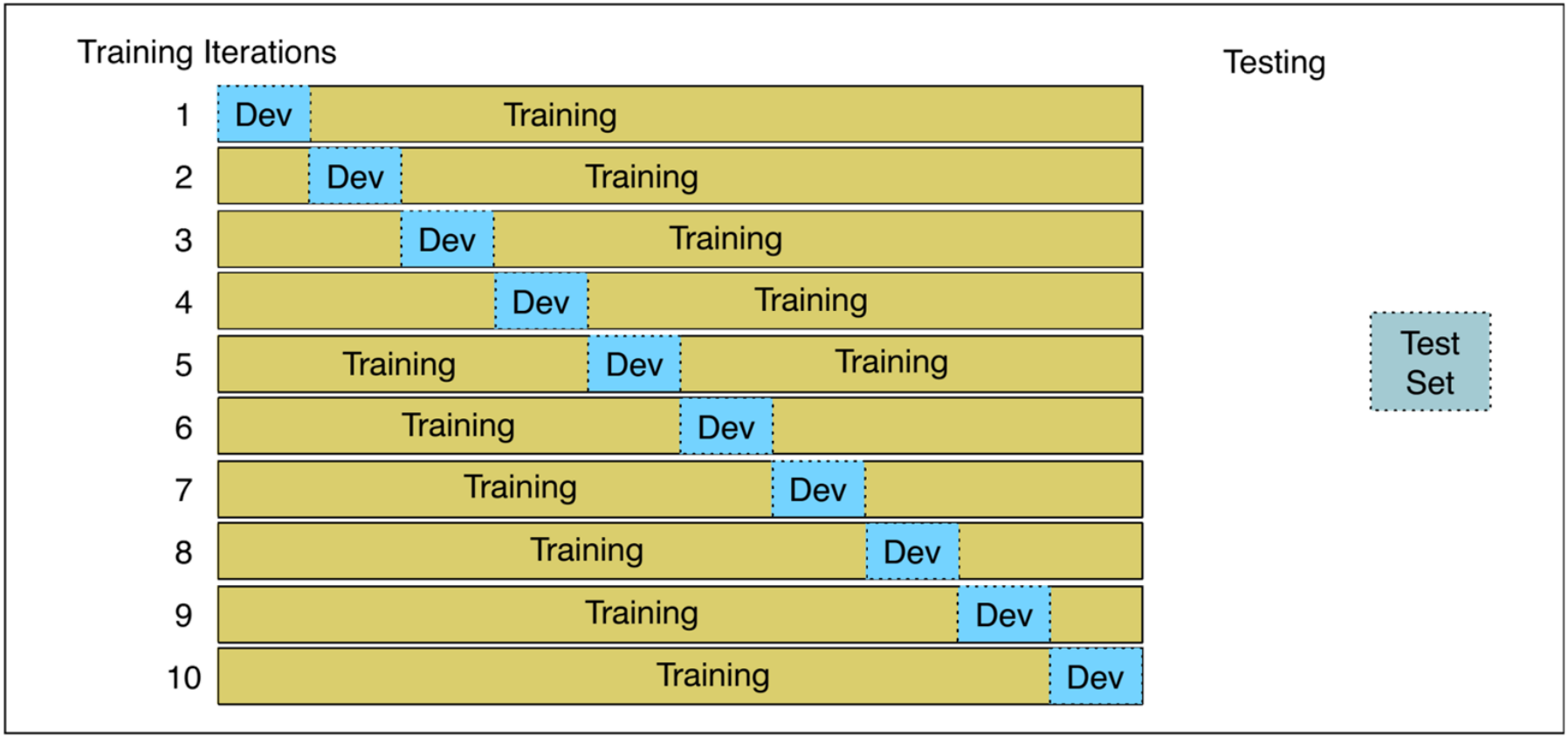


Figure 6.7 10-fold crossvalidation

Text Classification and Naïve Bayes

Text Classification: Practical Issues